

# Known Local Infrared Emission

Schedule one closed-catalog envelope without emitting light

E0 SYMBOLIC INTENT — NO EMITTER, CARRIER, PIN, OR TIMER

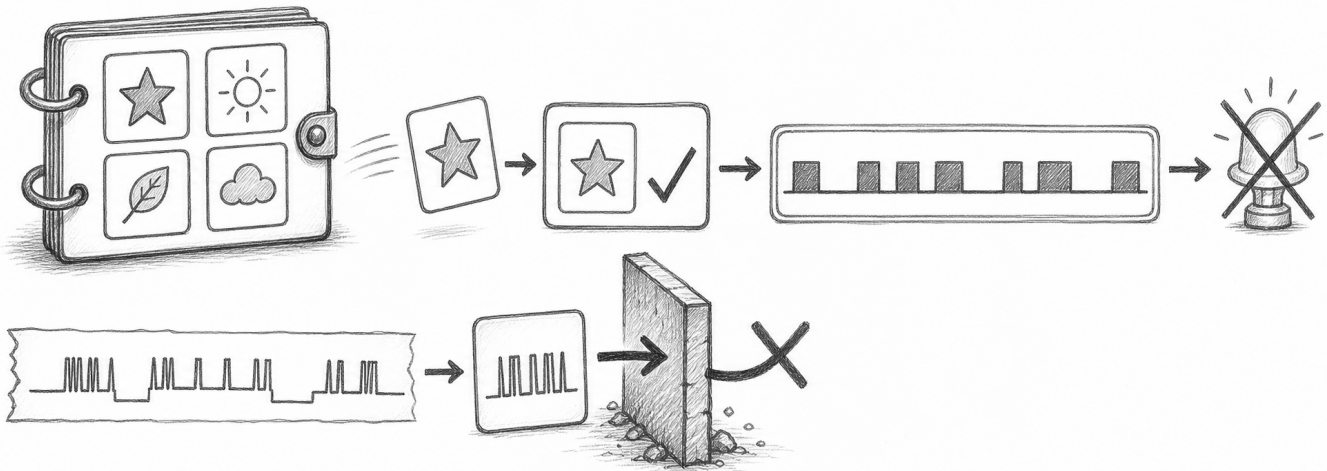
**Lesson** 053 — component  
**Prerequisite** explicit modular time  
**Evidence** named memory result cells

**Time** 110–150 minutes  
**Board** compile-only Mega replay  
**Boundary** immutable local catalog

**Responsibility.** `KnownIrEmissionPolicy` turns one firmware-authored symbolic command into bounded `CarrierOn`, `CarrierOff`, or `Inactive` memory intent. It owns no output, timer, pin, interrupt, carrier, emitter, receiver, supply, or optical path.

## 1 Select only from the closed catalog

The four harmless local symbols are `StationPing`, `StationReady`, `StationCancel`, and `StationAcknowledge`. The catalog revision and digest cover every semantic entry field. Runtime code cannot add, edit, import, learn, or infer an entry.



*Alternative text: a closed graphite catalog selects one symbol card, retains and checks its preview, and derives a bounded mark-space envelope ending inactive. A raw pulse strip below meets a solid barrier and cannot enter the catalog path.*

No public request accepts a pulse frame, captured view, raw address or command, duration array, waveform, or arbitrary bytes. Lesson 052 evidence therefore has no type-directed path to transmission.

## 2 Prepare, verify, then commit once

`prepare(codeId, transactionId, now)` reserves one candidate bound to the policy instance, configuration revision, epoch, policy and candidate generations, transaction ID, selected symbol, catalog identity, start and completion times, and candidate digest.

1. Prepare exactly one symbolic request; a second request reports busy.
2. Inspect the retained preview without mutating active policy.
3. Require `canCommit()` at the exact proposed start time.
4. Commit the unchanged preview once, or cancel it before commit.
5. Derive the envelope intent true at each supplied time directly.

Foreign, reconstructed, changed, stale, consumed, cancelled, or earlier-epoch previews reject without mutation. A delayed commit is not backdated or shifted; prepare a fresh candidate for a later start.

### 3 Never catch up missed transitions

The policy does not replay carrier cycles. It computes the current envelope phase in constant work. A late update publishes only the intent true at that time. At or after completion it remains **Inactive**; no missed burst is compressed into the next call.

| Boundary             | Prediction                | Interpretation                |
|----------------------|---------------------------|-------------------------------|
| exact start          | first envelope intent     | not electrical carrier start  |
| mark/space edge      | one current intent        | no loop over prior edges      |
| repeated time        | byte-identical state      | no duplicate transition       |
| late service         | current phase or complete | no catch-up burst             |
| exact completion     | inactive, complete        | not proof of optical darkness |
| half-range ambiguity | reject atomically         | never guess ordering          |

### 4 Make cancellation dominant

Before commit, exact candidate cancellation records **CancelledBeforeCommit** and never publishes **CarrierOn**. After commit, cancellation by the exact transaction publishes **Inactive** immediately and records **CancelledActive**. A matching repeated terminal cancellation is idempotent.

At a shared timestamp, a composing coordinator consumes cancellation before commit, update, or application of intent. Direct callers that split those inputs cannot claim call-order-independent dominance. Shutdown also publishes inactive intent and advances generation so old previews cannot revive.

## 5 Run the secret-handshake workbook

1. Initialize. Predict idle lifecycle, inactive intent, immutable catalog revision, and catalog digest.
2. Prepare **StationPing**. Try a second prepare, foreign preview, changed preview, stale generation, and a delayed commit.
3. Commit at the exact start. Sample every representative mark and space boundary without iterating missed intervals.
4. Cancel a prepared request, then cancel an active request. Collide cancellation with a due transition and give cancellation precedence.
5. Update after completion and verify that no catch-up burst appears.
6. Exercise timestamp rollover, regression, and half-range ambiguity.
7. Shutdown and reinitialize. Verify that the prior preview stays stale.

| time  | symbol | phase | envelope intent | transaction |
|-------|--------|-------|-----------------|-------------|
| _____ | _____  | _____ | _____           | _____       |
| _____ | _____  | _____ | _____           | _____       |
| _____ | _____  | _____ | _____           | _____       |

## 6 Separate acquisition evidence from safe-state evidence

**Acquisition prediction.** A supported configuration, nonzero epoch, and immutable catalog identity allow the pure policy to initialize.

**Acquisition observation.** Host tests inspect lifecycle, catalog identity, preview bindings, transaction attribution, phase, deadline, and envelope intent in memory.

**Acquisition interpretation.** Success proves closed-catalog transaction and timing semantics only. It does not acquire a timer, pin, emitter, driver, receiver, supply, or carrier endpoint.

**Safe-state prediction and observation.** Construction, failed initialization, cancellation, completion, fault, shutdown, and restart publish **Inactive** memory intent. That is not an emergency stop, power removal, or electrical proof that an emitter is dark.

## 7 Read the conservative resource evidence

A fresh final canonical Mega 2560 build measures 4,854 bytes of flash and 276 bytes of static SRAM with Arduino AVR core 1.8.8 and avr-gcc 7.3.0. `sizeof(KnownIrEmissionPolicy)` is 74 bytes. Conservative compiler stack evidence estimates about 155 bytes on the measured deepest analyzed path.

These values are compiler evidence for the exact enhanced example, source, toolchain, optimization, and fixture. They are not runtime high-water measurements, proof of every interrupt schedule, or evidence of a carrier or emitter. Re-measure after any source, core, compiler, flag, or composition change.

## 8 Keep E1 explicitly open

No exact emitter is selected, and the authorized kit inventory does not establish one. E1 must qualify the exact emitter, driver, resistor, supply, carrier timer/channel/pin, receiver, observation path, optical geometry, default-off behavior, and bench acceptance. Current limiting and bounded duty reduce exposure; they do not establish eye safety. This lesson makes no eye-safety claim and supplies no wiring or formal schematic.

## 9 Record the E0 result

- acquisition status: \_\_\_\_\_ catalog revision: \_\_\_\_\_
- prepared symbol: \_\_\_\_\_ transaction: \_\_\_\_\_
- completion intent: \_\_\_\_\_ cancellation result: \_\_\_\_\_
- safe-state result: \_\_\_\_\_

Lesson 054 may map one fixed valid receive symbol to a different fixed local symbol. Repeat, unknown, malformed, self-echo, and arbitrary captured data remain ineligible.